

Khulna University, Khulna
Electronics and Communication Engineering Discipline
4th Year Term I Examination, Session: 2021-2022
Course No.: ECE 4123
Full Title of the Course: Control Systems
Full Marks: 60 Time: 03 Hours

- The figure in the margin indicates full marks. The questions are of equal value.
- Use separate sheet for each section.

SECTION A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) Mention some advantages and disadvantages of open-loop and closed-loop control systems with proper examples. 02
- (b) Describe the importance of control systems in electronics and communication engineering field. 02
- (c) Draw the block diagram of an insulin delivery control system for open-loop and closed-loop control of blood glucose. Briefly explain the operation of the system. 06
2. (a) Briefly describe three approaches to the mathematical modeling of control systems. Draw the free-body-diagram of a quarter car suspension system with two-degree-of-freedom. 05
- (b) Find out the transfer function for the proportional controller shown in Fig. 2(b). 03

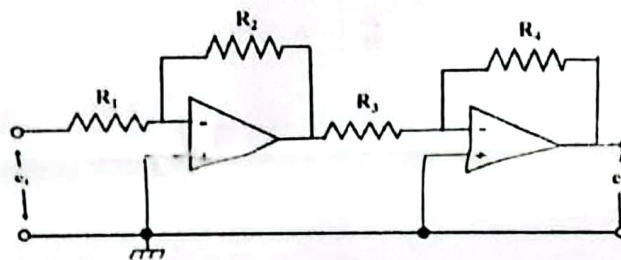


Fig. 2(b)

- (c) Describe what is meant by analogous system. Sketch a mechanical analog of a series R-L circuit. 02
3. (a) Given a control system represented in state space model by the following equations: 07

$$\dot{x} = \begin{bmatrix} 0 & 2 \\ -2 & -5 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-2t}$$

$$y = [2 \quad 1] x, \quad x(0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Do the following: (i) Solve for the state transition matrix.

 (ii) Solve for the state vector using the convolution integral.

 (iii) Find the output, $y(t)$.

- (b) Determine the impulse response, step response and ramp response for the system shown in Fig. 3(b). 03

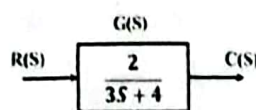


Fig. 3(b)

- 4.(a) Briefly describe the typical response of a non-minimum phase system with proper diagram. 02
- (b) Using Mason's rule, find the transfer function for the system responded by Fig. 4(b). 04

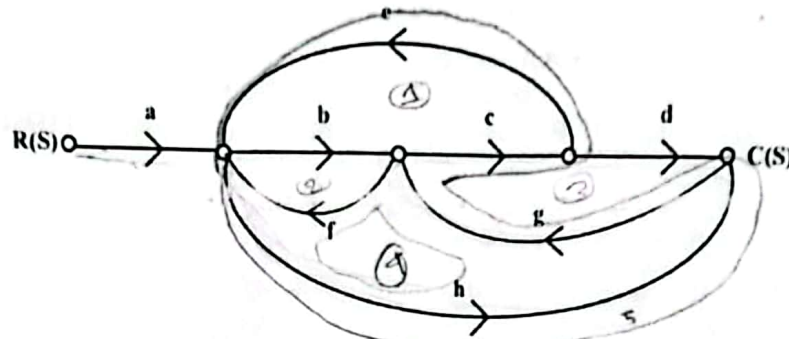


Fig. 4(b)

- (c) The block diagram of an electronic pacemaker is given in Fig. 4(c). What is the type of this system? Determine the value of K for which the steady state error to a unit ramp input will be 0.01. 04

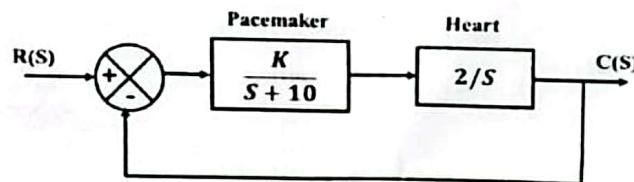


Fig. 4(c)

SECTION B

There are **FOUR** questions in this section. Answer any **THREE** questions.

- 5.(a) Apply R-H criteria to determine the stability of the system shown in Fig. 5(a). 05

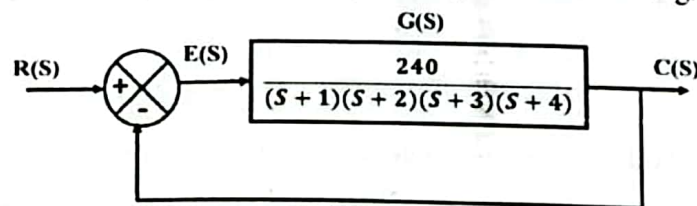


Fig. 5(a)

- (b) The closed loop transfer function of a system is 05

$$\frac{C(s)}{R(s)} = \frac{K}{S(S^2 + S + 1)(S + 4) + K}$$

Determine the range of K for which frequency the system will cause self-sustained oscillation?

6. (a) What is a root locus? How can you tell from the root locus if a system is unstable? 02

(b) Given a unity feedback system that has the forward transfer function $G(s) = \frac{K(s+2)}{(s^2-4s+13)}$ 08

Do the following:

- (i) Sketch the root locus, (ii) Find the imaginary axis crossing, (iii) Find the gain, K , at $j\omega$ -axis crossing, (iv) Find the break in point, (v) Find the angle of departure from the complex poles.

7. (a) Draw the Bode magnitude plot of the transfer function 06

$$G(j\omega) = \frac{4(1 + j\omega/2)}{j\omega(1 + \frac{j\omega}{10} - \frac{\omega^2}{100})}$$

(b) Consider a feedback system whose open loop transfer function is given by 04

$$G(s)H(s) = \frac{2}{s(s^2 + 0.8s + 2)}$$

Investigate the stability of the system using Nyquist plot.

8. (a) Determine the gain margin and phase margin of an unity feedback system having an open loop transfer function 05

$$G(j\omega) = \frac{10}{j\omega(j0.1\omega + 1)(j0.05\omega + 1)}$$

(b) Formulate the transfer function of the electric lag network shown below: 05

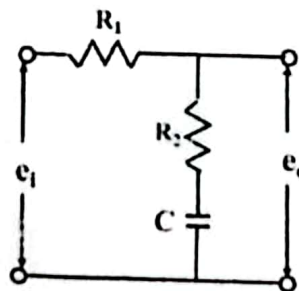


Fig. 8(b)

Also, draw the S plane representation of the lag compensator.

253

0.67 + 0.33

- The figure in the margin indicates full marks. The questions are of equal value.
- Use separate sheet for each section.

SECTION A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) What is division point? Justify its significance in ammeter voltmeter method. 02
- (b) Explain why do we need to measure the earth resistance. Describe the fall of potential method for the measurements of earth resistance. 04
- (c) In a Carey-Foster slide wire bridge a resistance of 1.0125Ω is compared with a resistance of 1Ω , the slide wire has a resistance of 0.025Ω in 100 divisions. The ratio arms are 10.05Ω and 9.95Ω , respectively. If the slide wire is 120 cm long, how far the balance position is found in scale divisions? 04
2. (a) Justify why Maxwell inductance capacitance bridge is useful for measuring of $1 < Q < 10$. If $Q > 10$, which bridge will be more suitable and why? 04
- (b) How the correction of air flux is done for the measurement of flux density in a ring specimen? Why is it needed? 03
- (c) In a Maxwell inductance capacitance bridge at balance condition if $R_1 = 240\Omega$, $R_3 = 600\Omega$, $R_4 = 1000\Omega$, $C_4 = 0.5\text{ pF}$, and $f = 1\text{ kHz}$, then determine the value of R_2 , L_1 and storage factor. 03
3. (a) Describe how a potential divider arrangement is used for the design of a multirange voltmeter. 03
- (b) What do you mean by loading effect of a voltmeter? Derive the expression of half scale resistance in a shunt type ohmmeter. 04
- (c) Design a multirange d.c. voltmeter using a basic meter movement with an internal resistance $R_m = 100\Omega$ and a full-scale deflection current $I_m = 1\text{ mA}$. The required ranges are $0 - 10\text{ V}$, $0 - 50\text{ V}$, $0 - 250\text{ V}$ and $0 - 500\text{ V}$. 03
4. (a) What is CRO? Explain electrostatic focusing technique used in CRO with necessary figure. 03
- (b) Define Lissajous pattern. How an unknown frequency and phase measurement can be performed using Lissajous pattern? Justify with necessary diagram and example. 04
- (c) A solenoid of 1 m long and wound with 900 turns has a search coil of 50 turns and cross-sectional area 350 mm^2 at its centre. Reversing a current of 5 A in the solenoid causes a deflection of 5 divisions in a ballistic galvanometer connected to the search coil. Determine the galvanometer constant in flux linkages per division and flux linkage sensitivity. 03

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Electronics and Communication Engineering Discipline

SECTION B

There are **FOUR** questions in this section. Answer any **THREE** questions.

- 5.(a) What is transducer? Justify how an inductive transducer can be used as a displacement transducer. 03
- (b) Define POT. Briefly explain the input and transfer characteristic of a transducer. 03
- (c) Why thermistor is called an ideal temperature transducer than platinum? 01
- (d) A parallel-plate capacitive transducer has a plate area of $40\text{mm} \times 40\text{mm}$ and plate spacing of 0.5mm . Calculate the capacitance and the displacement that causes the capacitance to change by 7pF . Also, find out the transducer sensitivity. 03
- 6.(a) What is the basic principle of temperature measurement using thermocouple? Briefly explain how the piezoelectric and oscillation transducers are used for the measurement of pressure. 04
- (b) Write down the advantages and disadvantages of turbine meter and electromagnetic flow meter. 03
- (c) For an instrumentation amplifier show that to increase the value of gain, the value of gain setting resistor needs to be decreased. 03
- 7.(a) What do you mean by aperture time? Justify in a D/A converter, the analog output voltage is proportional to the digital input. 03
- (b) Define modulus-16 counter. Describe the operation of $3^{1/2}$ -digit display counter with necessary figure. 04
- (c) An analog transducer with a $0 - 10\text{V}$ input is able to distinguish a change of 10mV in its input signal. Calculate its resolution, the number of bits of an A/D converter that uses a binary code and has almost the same resolution as the transducer, the quantization error and the number of decision levels. 03
- 8.(a) Why dual-slope-integrator DVM is preferred over ramp-type DVM? Describe the operation of digital frequency meter with necessary figure. 04
- (b) Define telemetry. Briefly explain the basic block diagram of pulse-code modulation telemetry system. 04
- (c) What are the main components of a digital data acquisition system? Mention some functions of this system. 02

$$1.75 \times 10^{-9}$$

$$0.175$$

Khulna University, Khulna
Electronics and Communication Engineering Discipline
4th Year, Term I Examination
Session: 2021-2022
Course No.: ECE 4107
Full Title of Course: Telecommunication Engineering
Full Marks: 60 Time: 03 Hours

- * The figures in the margin indicate full marks. The questions are of equal value.
- * Use separate sheet for each section.

Section A

There are **Four** questions in this section. Answer any **Three** questions.

- | | | |
|-------|---|----|
| 1.(a) | Describe the basic approaches to the design of subscriber access to a Strowger systems. | 04 |
| 1.(b) | Design a 100-line switching system using Uniselector. | 04 |
| 1.(c) | Explain the use of side tone in telecommunication. | 02 |
| | | |
| 2.(a) | Discuss how a carbon granule microphone acts as an amplitude modulator. | 04 |
| 2.(b) | Identify the necessity of keeping the magnetic diaphragm in an earphone displaced from its <u>unstressed position</u> ? How is this achieved? | 03 |
| 2.(c) | A 100-line exchange is partly folded and non-folded. Forty percent of the subscribers are active during the peak hour. If the ratio of local to external traffic is 5:1, estimate the number of trunk lines required. | 03 |
| | | |
| 3.(a) | Describe how a uni-selector can be used as a selector hunter. | 05 |
| 3.(b) | Calculate the design parameters such as N , S , SC , C , TC , EU , P_B , CCI , and K of a 5000 line exchange where switch system consists one uni-selector for each subscriber and all these uni-selectors share 1200 two-motion selectors. Assume the price of each uni-selector and two-motion selector are \$10 and \$25, respectively. Also, cost of the common hardware is around \$15000. | 05 |
| | | |
| 4.(a) | Discuss briefly the centralized and distributed SPC with proper block diagram. | 04 |
| 4.(b) | Determine the implementation complexity of a 131072 channel TSSST switch designed to provide a maximum blocking probability of 0.002 under channel occupancies of 0.7. Assume the switch services 1024 TDM input links with 128 channels on each link. | 03 |
| 4.(c) | Explain the basic function of step-by-step switching system. | 03 |

Section B

There are **Four** questions in this section. Answer any **Three** questions.

- | | | |
|-------|--|----|
| 5.(a) | Define the term traffic and call completion rate. Differentiate BHCA and BHCR. | 05 |
| 5.(b) | Explain why grade of service and blocking probability are important teletraffic parameters. | 05 |
| 6.(a) | In a telephone system, there are 20 servers and 100 subscribers. On an average there are 10 busy servers at any time. The probability of all the servers being busy is 0.2. Calculate the grade of service for both Erlang and Engset traffic. | 04 |
| 6.(b) | Over a 20-minute observation interval, 40 subscribers initiate calls. Total duration of the calls is 4800 seconds. Calculate the load offered to the network by the subscribers and the average subscriber traffic. | 03 |
| 6.(c) | Explain briefly different models of loss system. | 03 |
| 7.(a) | Differentiate the different signaling techniques (to be done) | 03 |
| 7.(b) | Explain common channel signaling in details. | 04 |
| 7.(c) | Define Fill-in Signal Unit (FISU). | 03 |
| 8.(a) | Define ISDN. Discuss the motivations behind ISDN. Mention few new services for ISDN. | 03 |
| 8.(b) | Discuss X.400 in the context of OSI model. Mention some features of X.400 services. | 03 |
| 8.(c) | Show that for an integrated service model: | 04 |

$$T_I = \frac{1}{\mu - \lambda}$$

Where the symbols have their usual meaning.

Khulna University
Electronics and Communication Engineering Discipline
4th Year Term I Examination 2023, Session: 2021-2022
Course No.: ECE 4103
Course Title: VLSI Circuits & Design
Full Marks: 60 Time: 3.00 Hours

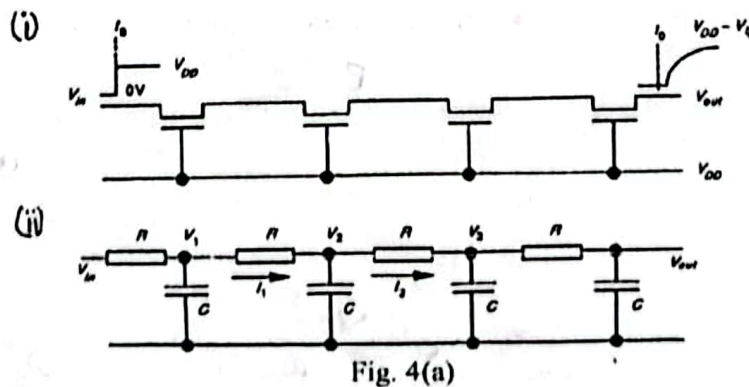
* The figures in the margin indicate full marks. The questions are of equal value.

* Use separate sheet for each section

Section A

There are four (04) questions in this section. Answer any three (03) questions.

- 1.(a) Sketch and explain the CMOS n-well fabrication steps. 04
- (b) Derive the required ratio between Z_{pu} and Z_{pd} if an nMOS inverter is to be driven from another nMOS inverter. 04
- (c) Explain the effect of channel length and width on transconductance for nMOS transistor. 02
- 2.(a) Formulate the following expression for a PMOS transistor 05
 - (i) $I_{sd} = \frac{\epsilon \mu_p W}{DL} \left[(V_{sg} - V_t) V_{sd} - \frac{V_{sd}^2}{2} \right]$
 - (ii) $I_{sd} = \frac{\epsilon \mu_p W}{2DL} (V_{sg} - V_t)^2$
- (b) Determine the drain to source current for both saturated and non-saturated region of a typical nMOS transistor where relative permittivity of insulating layer is 4. Channel length, width and insulating layer thickness are $1 \mu m$, $2 \mu m$, and $3 \mu m$, respectively. Also, $V_{gs} = 0.8V_{DD}$, $\mu_n = 650 cm^2/V$ and the symbols V_{DD} and V_t have their usual value. 05
- 3.(a) Draw the stick diagram and mask layout of the following expressions: 04
 - (i) $Y = \overline{A + B + C}$ use nMOS, and (ii) $Y = \overline{X \cdot Z}$ use CMOS
- (b) For a CMOS inverter, calculate the shift in the transfer characteristics when the β_n/β_p ratio is varied from 1/1 to 10/1. 03
- (c) A particular layer of MOS circuit has a resistivity $\rho = 1 \text{ ohm cm}$. A section of this layer is $55 \mu m$ long and $5 \mu m$ wide and has a thickness of $1 \mu m$. Calculate the resistance from one end of this section to the other. What is the value of Rs ? 03
- 4.(a) Analyse Fig. 4(a) and formulate the expression $\tau_d = n^2 r c (\tau)$ where the symbols have their usual meaning. 05



Date: 07-08-22

Khulna University, Khulna
Electronics and Communication Engineering Discipline
4th Year 1st Term Examination
Session: 2021-2022
Course No.: ECE 4101
Full Title of the Course: Industrial Electronics
Full Marks: 60 Time: 03 Hours

- The figure in the margin indicates full marks. The questions are of equal value.
- Use separate sheet for each section.

SECTION A

There are **FOUR** questions in this section. Answer any **THREE** questions.

- 1.(a) Draw the $V-I$ characteristics of an SCR. Describe the operating principle of an SCR with gate triggering. 03
- 1.(b) What is meant by commutation? Illustrate external pulse commutation technique with suitable diagram. 03
- 1.(c) What is the purpose of series and parallel connections of SCR. Justify the necessity "when several SCRs are required to operate in parallel, it should be ensured that they operate precisely at the same temperature". 04
- 2.(a) Draw the basic structure of an UJT and illustrate the characteristics of cut-off, negative resistance and saturation region of the device. 04
- 2.(b) In a data sheet, intrinsic stand-off ratio (η) of an UJT is given from 0.74 to 0.86. Evaluate the maximum and minimum peak-point voltage for the UJT with $V_{BB} = 25$ V. 02
- 2.(c) Explain the working principle of DIAC and TRIAC based AC phase control circuit. Also draw the output waveforms of the phase control circuit. 04
- 3.(a) How the power handling capability of a power diode is increased than normal PN diode? Illustrate with necessary diagram. 03
- 3.(b) For a punch-through power diode, deduce the following relation: 04
- $$V_{BD} = E_{BD}W_d - \frac{qN_dW_d^2}{2\epsilon},$$
- Where symbols have their usual meaning.
- 3.(c) Do you expect any difference in output waveform, if a rectifier circuit is designed using a power diode or a Schottky diode operating at 50 kHz? If yes, why? 03
- ✗ 4.(a) Draw the N-channel VDMOS power MOSFET and illustrate its working principle. 04
- ✓ 4.(b) In an SCR half-wave rectifier circuit, the measured average load current is 1 A at a firing angle of 30° . Calculate peak-load current. 02
- ✗ 4.(c) A circuit is connected as shown in Figure 4 (c). Neglect the diode voltage drop and determine the current waveform, the average load voltage and the average load current of (a) a pure resistor of 10Ω , (b) An inductance of 0.1 H in series with a 10Ω resistor. Assume the extinction angle $\beta = 265^\circ$. 04

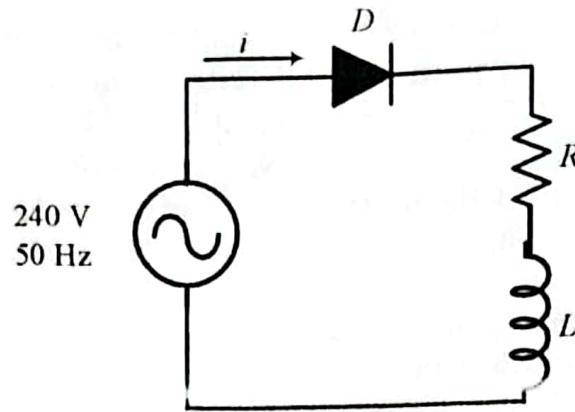
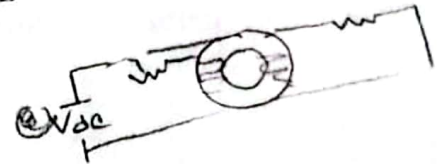


Figure 4(c).



SECTION B

There are **FOUR** questions in this section. Answer any **THREE** questions.

- 5.(a) Define stepper motor. What are the types of stepper motor? 02
- 5.(b) Explain the working principle of a four-step permanent magnet stepper motor with necessary diagram. Also, show the switching sequence for a four-step permanent magnet stepper motor. 05
- 5.(c) Describe the operating principle of a bipolar stepper motor with necessary diagram. 03
- 6.(a) Define inverter. Explain the operation of a single-phase half-bridge voltage source inverter (VSI) with necessary diagrams. Show the switching states of the single-phase half-bridge VSI. Also, draw the necessary waveforms to represent the sinusoidal pulse-width-modulation (SPWM) using the single-phase half-bridge VSI. 06
- 6.(b) Describe the operation of a Buck chopper with the circuit diagram. Also draw the waveforms of output voltage and current for a Buck chopper circuit. 04
- 7.(a) Explain the working principle of an induction motor with necessary diagram. 03
- 7.(b) Describe the working principle of a cycloconverter with necessary diagram. 04
- 7.(c) A three-phase, 440 V, 50 Hz star connected, 4 pole squirrel cage induction motor has the following equivalent circuit parameters: $R_s = 0.3 \Omega$, $R_r = 0.2 \Omega$, $X_s = X_r = 0.4 \Omega$, $X_m = 20 \Omega$. Rotor speed = 1440 rpm. Compute (i) synchronous speed N_{os} , (ii) slip s , and (iii) input impedance of the induction motor. 03
- 8.(a) Draw the circuit diagram of a magnetic amplifier and explain its operation. 03
- 8.(b) Explain seam welding with necessary diagram. List the application areas of seam welding. 03
- 8.(c) Define induction heating. For induction heating show that, the power entering per square centimeter of surface is given by $P_t = \frac{8\pi H_0^2 \sqrt{10^{-9} \mu_r f}}{\sqrt{\sigma}}$, where the symbols have their usual meaning. 04

mech

mech

~ 2 ~

1208